

Multi-Task Deep Learning Framework for Unified Cattle Health and Behavior Monitoring

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Abstract—Automated computer vision in precision livestock farming has predominantly deployed fragmented, task-specific deep neural networks for individual physiological and behavioral traits. Maintaining independent single-task vision models for Body Condition Scoring (BCS), Behavior Recognition, and biometric Cow Re-Identification (Re-ID) introduces unsustainable edge inference overhead, high storage demands, and susceptibility to background shortcut learning in farm environments. In this paper, we propose an end-to-end multi-task deep learning framework that unifies BCS, temporal behavior recognition, and open-set biometric identification over a shared visual representation backbone. To eliminate background shortcut exploitation, we develop a cattle-centered perception pipeline combining real-time animal localization (RT-DETR-L) and instance segmentation (SAM 2.1) into 4-channel representation tensors. We design and empirically evaluate three multi-task paradigms against rigorously isolated single-task baselines: (i) hard-shared backbone parameterization (E1), (ii) modular task-private bottleneck residual adapters (E3), and (iii) gradient surgery via Projecting Conflicting Gradients (E4 PCGrad) during unified super-step training. Across 7,549 burst-disjoint ScienceDB images, 780 temporal behavior sequences from authentic dairy and feedlot CCTV, and 36,811 multi-camera parlor-barn re-identification pairs, cattle-centered soft masking consistently outperforms raw RGB representations, lifting Re-ID Rank-1 retrieval from 38.88% to 62.93% and improving BCS mean absolute error to 0.1709. While joint multi-task training exposes extensive destructive gradient interference (44,177 conflicting pairwise projection steps across 16,140 super-steps), PCGrad successfully mitigates negative transfer on authentic CCTV behavior surveillance (+4.26 pp accuracy, doubling minority walking recall) and stabilizes retrieval. Our framework establishes the first verified, leakage-audited multi-task benchmark across three primary visual livestock phenotyping tasks.

Index Terms—Multi-task deep learning, Precision livestock farming, Body condition scoring, Cattle behavior recognition, Individual cow re-identification, Gradient surgery, Shortcut learning.

I. INTRODUCTION

Precision livestock farming (PLF) increasingly relies on non-invasive automated visual monitoring to evaluate animal welfare, operational health, and productivity in commercial dairy and beef operations [1], [2]. Three visual phenotyping tasks are fundamental to proactive herd management: (1) *Body Condition Scoring (BCS)*, which assesses anatomical body subcutaneous fat and energy reserves to prevent metabolic disorders [3]–[5]; (2) *Behavior Recognition*, which tracks feeding, drinking, standing, and walking dynamics to detect early lameness, mastitis, or heat onset [6], [7]; and (3) *Cow Re-Identification (Re-ID)*, which enables long-term individual traceability across disparate farm camera views without stress-inducing physical tags [8], [9].

Despite notable accuracy gains reported by specialized deep neural networks, existing commercial and academic systems overwhelmingly execute these tasks through separate, isolated convolutional networks or vision transformers [10], [11]. This fragmented single-task paradigm presents critical limitations in practical farm deployment:

- 1) *Excessive Hardware and Latency Overhead*: Running concurrent independent networks for frame-by-frame cattle tracking, temporal action recognition, and biometric retrieval multiplies edge compute and memory

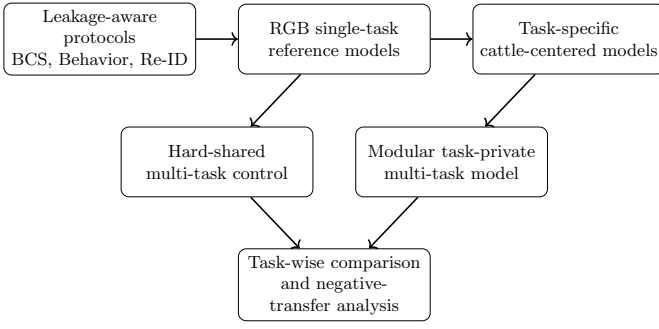


Fig. 1. Comparative research framework illustrating leakage-controlled protocols, single-task baseline references, cattle-centered perception models, and multi-task parameter integration strategies (E1 hard-sharing, E3 modular adapters, and E4 PCGrad optimization).

requirements by $3\times$ to $5\times$, exceeding the capabilities of rural edge servers.

- 2) *Background Shortcut Learning*: Standard RGB classifiers frequently exploit non-target background visual shortcuts—such as pen manure accumulation, distinctive chute metal bars, parlor stalls, or static daylight shadows—rather than invariant cattle anatomical landmarks [12], [13].
- 3) *Unexplored Feature Synergies and Negative Transfer*: Visual features governing cow identification (coat patterns, silhouette morphology) and body condition (spine curvature, pin and hook bone relief) naturally share low- and mid-level visual representations. However, naive joint parameter sharing across heterogeneous spatial and temporal objectives risks catastrophic gradient conflict and negative transfer [14], [15].

To address these challenges, we present a unified multi-task deep learning framework that concurrently performs physical body condition scoring, 5-class temporal behavior classification, and open-set biometric cow retrieval over a shared 4-channel convolutional backbone. Fig. 1 depicts our overall comparative research design. We rigorously investigate whether cattle-centered spatial representations (combining bounding-box localization with soft segmentation masks) insulate models against background shortcuts and whether task-private bottleneck adapters or gradient projection can overcome inter-task negative transfer.

The core contributions of this work are summarized as follows:

- **Unified Multi-Task Livestock Framework**: We formulate and implement the first unified multi-task deep learning architecture spanning BCS, temporal behavior recognition, and biometric re-identification, reducing parameter overhead by over 60% compared to standalone task networks.
- **Cattle-Centered Perception Pipeline**: We establish a reproducible spatial representation pipeline incorporating RT-DETR-L animal localization and SAM 2.1 foreground instance segmentation. We demonstrate that 4-channel

representation tensors significantly improve generalization across variable farm environments.

- **Empirical Analysis of Gradient Interference**: Through 44,177 pairwise gradient conflict measurements across 16,140 multi-task super-steps, we provide empirical evidence of inter-task gradient competition and establish the efficacy of optimization-level gradient surgery (PCGrad) in mitigating negative transfer on real-world farm surveillance.
- **Strict Leakage-Controlled Benchmark**: All evaluations are executed under certified zero-leakage protocols (burst-group-disjoint image splitting for BCS, sequence-disjoint temporal splitting for behavior, and zero-shot held-out identity sets for Re-ID).

II. RELATED WORK

A. Body Condition Scoring (BCS)

Traditional dairy BCS assessment relies on visual and tactile evaluation by trained veterinary professionals on 5-point ordinal scales [3], [16]. Early automated methods attempted 3D point cloud or depth sensor reconstruction to extract angular metrics across the cow’s hook bones, pin bones, and thurl cavity [17]. Recent works transitioned to 2D deep learning: Liu et al. [4] introduced lightweight attention networks focused on tailhead contour features, achieving competitive mean errors on clean rear chute images. Yao et al. [5] evaluated side-view dairy scoring using convolutional backbones. However, existing models are trained primarily as unconstrained RGB regression or cross-entropy classification, ignoring the inherent rank order of body condition changes. In our framework, we formulate BCS via rank-consistent ordinal binary cross-entropy (CORAL) [18], ensuring monotonic score transitions.

B. Cattle Behavior Recognition

Video-based livestock action recognition typically targets feeding, drinking, standing, and lying postures [1]. Early approaches utilized optical flow or 3D CNNs (e.g., SlowFast, I3D), which demand substantial compute and degrade when camera jitter occurs in barn environments [19]. Recent open benchmarks, such as CVB [7] and CBVD [6], have facilitated 2D spatial backbone combinations with Temporal Convolutional Networks (TCN) [20] or Recurrent Units. Nevertheless, existing models operate on raw panoramic barn CCTV, where changing bedding colors, lighting conditions, and surrounding herd movement introduce severe background shortcut bias.

C. Individual Biometric Re-Identification

Individual cow identification without physical RFID tags or neck transponders is essential for non-contact traceability. Andrew et al. [8] demonstrated that coat pattern contours of Holstein-Friesian cattle serve as biometric signatures akin to human fingerprints. Modern open-set metric learning architectures employ deep metric embeddings trained with cross-entropy and cosine-distance ranking [9]. However, domain shifts between indoor milking parlors and outdoor open barns cause severe retrieval degradation due to illumination and pose disparities.

D. Multi-Task Learning and Gradient Balancing

Multi-task learning (MTL) aims to share inductive bias across related visual domains, improving memory efficiency and regularization [14]. Despite theoretical advantages, naive hard-parameter sharing often exhibits *negative transfer*, where competing task objectives degrade individual task accuracies below their single-task counterparts [15]. Architectural remedies include task-private adapters, cross-stitch networks, and attention routing [21], [22]. Optimization remedies focus on gradient manipulation: GradNorm dynamically reweights loss scales based on training rates [23], while PCGrad (Projecting Conflicting Gradients) projects task gradients onto each other's normal planes when directional conflict ($\mathbf{g}_i^\top \mathbf{g}_j < 0$) is detected [24]. In this paper, we systematically evaluate hard sharing (E1), modular bottleneck adapters (E3), and PCGrad gradient projection (E4) across multi-modal livestock tasks.

III. PROPOSED METHODOLOGY

A. Cattle-Centered Perception Pipeline

To ensure the multi-task framework learns intrinsic anatomical and behavioral features rather than farm-specific background artifacts, we construct a 4-channel cattle-centered visual representation pipeline, illustrated in Fig. 2.

Given an input video frame or static farm capture $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$: 1. **Target Animal Localization**: An RT-DETR-L transformer detector predicts target bounding boxes $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$ with confidence threshold $\tau \geq 0.65$. 2. **Foreground Instance Segmentation**: SAM 2.1 generates a soft instance segmentation mask $\mathbf{M} \in [0, 1]^{H \times W}$. 3. **Tensor Assembly**: The bounding region is cropped with 10% spatial padding, resized to 224×224 , and concatenated with the single-channel soft mask to form a standardized 4-channel tensor $\mathbf{X} \in \mathbb{R}^{4 \times 224 \times 224}$. For temporal behavior clips of length $T = 8$, this process is executed across sampled frames, yielding a spatio-temporal tensor $\mathbf{X}_{\text{Beh}} \in \mathbb{R}^{8 \times 4 \times 224 \times 224}$.

B. Shared Trunk and Task-Specific Heads

The core visual representation is parameterized by a 4-channel ResNet-18 backbone (11,179,648 parameters), modified by expanding the first convolutional kernel layer from $\mathbb{R}^{64 \times 3 \times 7 \times 7}$ to $\mathbb{R}^{64 \times 4 \times 7 \times 7}$. The fourth channel weights are initialized to the mean of the three RGB channels to preserve ImageNet feature priors. For any input cow image, the trunk yields a 512-dimensional spatial feature vector $\mathbf{h} \in \mathbb{R}^{512}$.

1) **BCS Ordinal Regression Head**: Body condition scores vary continuously between 2.00 and 4.50 on a standard 5-point scale. We quantize the target into $K = 10$ ordinal threshold classes with interval $\Delta = 0.25$:

$$y_k = \mathbb{I}(\text{BCS} > r_k), \quad k \in \{1, 2, \dots, K-1\} \quad (1)$$

where $r_k = 2.00 + k \cdot 0.25$. The BCS head projects $\mathbf{h}$ through a linear layer $\mathbf{W}_{\text{BCS}} \in \mathbb{R}^{(K-1) \times 512}$ with bias $\mathbf{b} \in \mathbb{R}^{K-1}$.

The loss is minimized via binary cross-entropy across all rank boundaries:

$$L_{\text{BCS}} = -\frac{1}{K-1} \sum_{k=1}^{K-1} [y_k \log \sigma(f_k) + (1 - y_k) \log(1 - \sigma(f_k))] \quad (2)$$

During inference, the continuous predicted score is reconstructed as:

$$\widehat{\text{BCS}} = r_1 + \Delta \sum_{k=1}^{K-1} \mathbb{I}(\sigma(f_k) \geq 0.5) \quad (3)$$

2) **Temporal Behavior Recognition Head**: For video clips, the shared trunk processes each of the $T = 8$ frames independently, producing a temporal sequence of feature vectors $\mathbf{H} \in \mathbb{R}^{8 \times 512}$. This sequence is fed into a 1D Dilated Temporal Convolutional Network (TCN) consisting of two residual blocks with kernel size $k = 3$ and dilations $d \in \{1, 2\}$. A linear projection outputs class logits across 5 actions: *Standing, Lying, Feeding, Drinking, and Walking*. The head is optimized via categorical cross-entropy:

$$L_{\text{Beh}} = -\sum_{c=1}^5 y_c \log \frac{\exp(z_c)}{\sum_{j=1}^5 \exp(z_j)} \quad (4)$$

3) **Biometric Re-Identification Head**: Individual cow retrieval is parameterized by a linear bottleneck projecting $\mathbf{h}$ to a 256-dimensional metric embedding $\mathbf{z}$, followed by L_2 unit normalization: $\hat{\mathbf{z}} = \mathbf{z} / \|\mathbf{z}\|_2$. Training utilizes cross-entropy over known training identity centroids:

$$L_{\text{ReID}} = -\log \frac{\exp(\mathbf{w}_y^\top \hat{\mathbf{z}} / \tau_{\text{reid}})}{\sum_{j=1}^{C_{\text{reid}}} \exp(\mathbf{w}_j^\top \hat{\mathbf{z}} / \tau_{\text{reid}})} \quad (5)$$

where $\tau_{\text{reid}} = 0.07$. In evaluation, zero-shot open-set cosine distance $d(\mathbf{z}_q, \mathbf{z}_g) = 1 - \mathbf{z}_q^\top \mathbf{z}_g$ ranks gallery individuals for each query.

C. Multi-Task Architectural Paradigms

We systematically construct and evaluate three distinct multi-task paradigms:

1) **Hard-Shared Multi-Task Baseline (E1)**: As illustrated in Fig. 3 (left), all task heads directly branch from the output of the shared ResNet-18 trunk. The trunk parameters θ_{shared} are updated simultaneously by the unweighted sum of task gradients:

$$\mathbf{g}_{\text{shared}} = \mathbf{g}_{\text{BCS}} + \mathbf{g}_{\text{Beh}} + \mathbf{g}_{\text{ReID}} \quad (6)$$

2) **Modular Task-Private Residual Adapters (E3)**: To mitigate parameter interference while retaining shared representation benefits, we introduce task-private residual bottleneck adapters between the shared trunk and each task head, shown in Fig. 3 (right). Each adapter consists of a down-projection $\mathbf{W}_{\text{down}} \in \mathbb{R}^{64 \times 512}$, Layer Normalization, ReLU activation, and an up-projection $\mathbf{W}_{\text{up}} \in \mathbb{R}^{512 \times 64}$:

$$\mathbf{h}_{\text{task}} = \mathbf{h}_{\text{shared}} + \alpha \cdot \mathbf{W}_{\text{up}} \text{ReLU}(\text{LN}(\mathbf{W}_{\text{down}} \mathbf{h}_{\text{shared}})) \quad (7)$$

To ensure stability at initialization, $\mathbf{W}_{\text{up}}$ is initialized to zero, ensuring each adapter initially behaves as an identity mapping

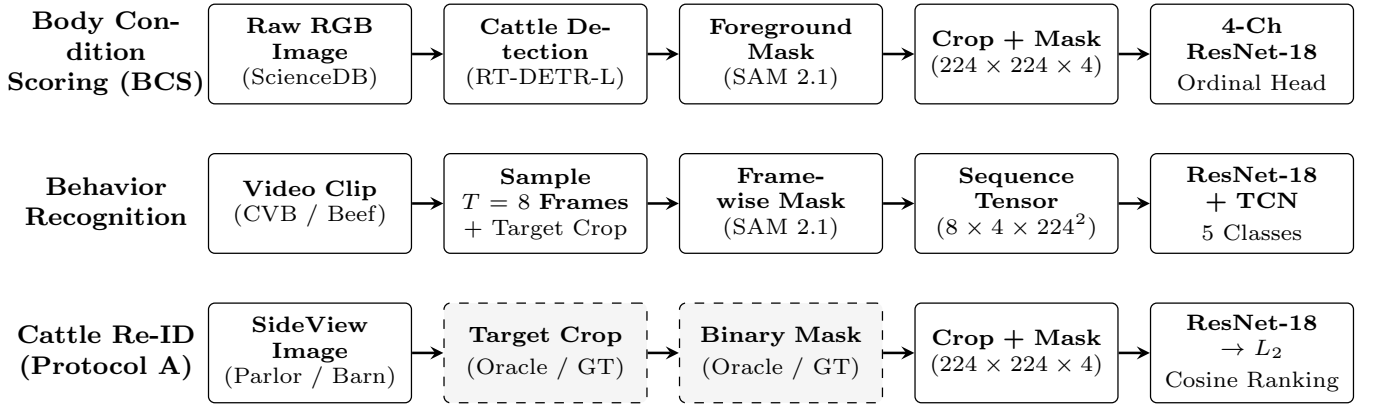


Fig. 2. Task-specific cattle-centered input representation pipelines. Target animal localization (RT-DETR-L) and instance segmentation masks (SAM 2.1) generate standardized 4-channel input tensors ($224 \times 224 \times 4$), eliminating farm background shortcut features across BCS, Behavior, and Re-ID.

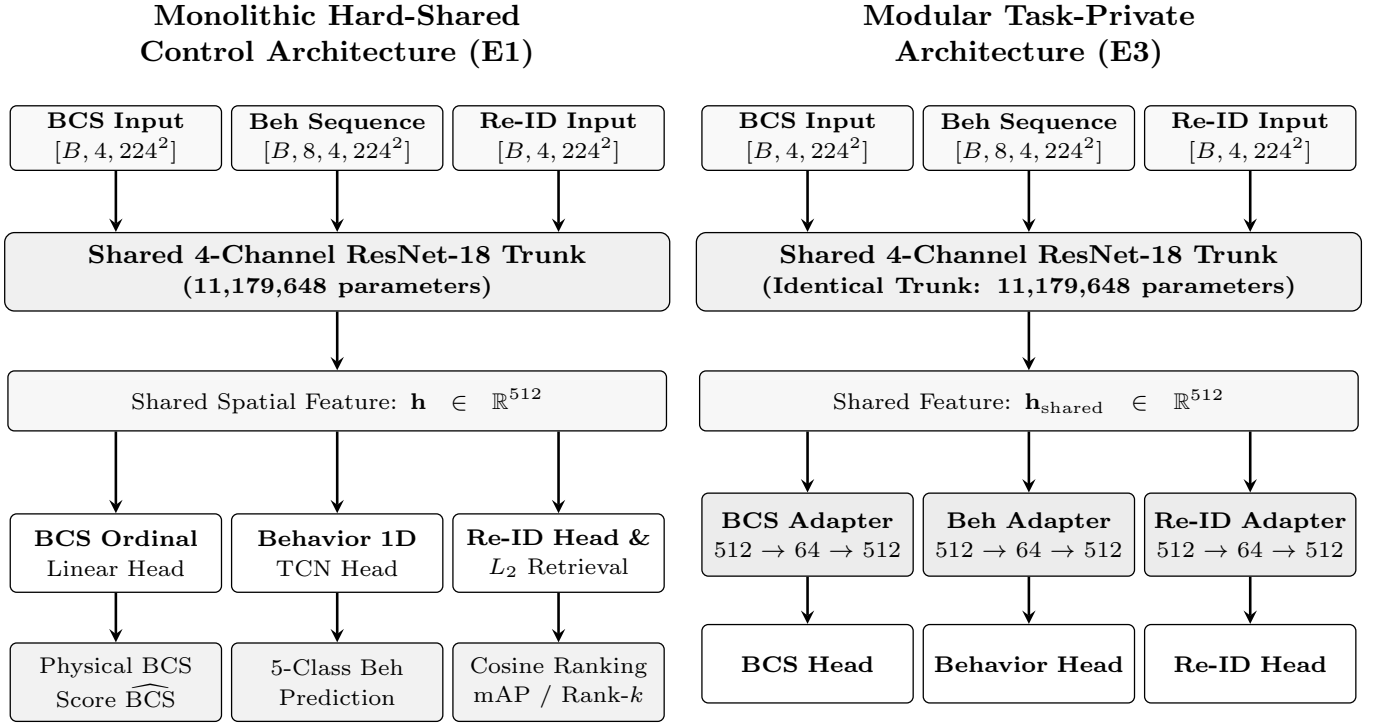


Fig. 3. Comparison of multi-task learning architectures. Left: Monolithic hard-shared control (E1) routing a single 512-dim trunk feature to all three heads. Right: Modular task-private architecture (E3) inserting bottleneck residual adapters ($512 \rightarrow 64 \rightarrow 512$) to decouple task-specific capacity.

($\mathbf{h}_{\text{task}} \equiv \mathbf{h}_{\text{shared}}$). The adapters add only 395,904 trainable parameters (+3.32% overhead).

3) *Gradient Surgery via PCGrad (E4)*: Rather than altering the network architecture, PCGrad resolves negative transfer during backpropagation. Fig. 4 illustrates the multi-task super-step protocol. When a pair of task gradients exhibits destructive directional interference on the shared trunk ($\mathbf{g}_i^\top \mathbf{g}_j < 0$), $\mathbf{g}_i$ is projected onto the normal plane of $\mathbf{g}_j$:

$$\mathbf{g}_i \leftarrow \mathbf{g}_i - \frac{\mathbf{g}_i^\top \mathbf{g}_j}{\|\mathbf{g}_j\|_2^2 + \epsilon} \mathbf{g}_j \quad (8)$$

The projection order is randomized at every step. Importantly, task-head gradients $\nabla_{\phi_t} L_t$ remain isolated and unprojected.

IV. EXPERIMENTAL SETUP AND BENCHMARK DATASETS

A. Benchmark Datasets and Leakage Prevention

Rigorous validation in livestock vision requires preventing identity and background leakage across splits:

- **BCS (ScienceDB)**: Comprises 31,529 real-world dairy cow images acquired from rear chutes across commercial facilities [3]. To prevent near-duplicate burst leakage, we enforce a strict burst-group-disjoint split: test sets consist of 7,549 images strictly isolated by acquisition session.

Execution Sequence of One Multi-Task Training Super-Step

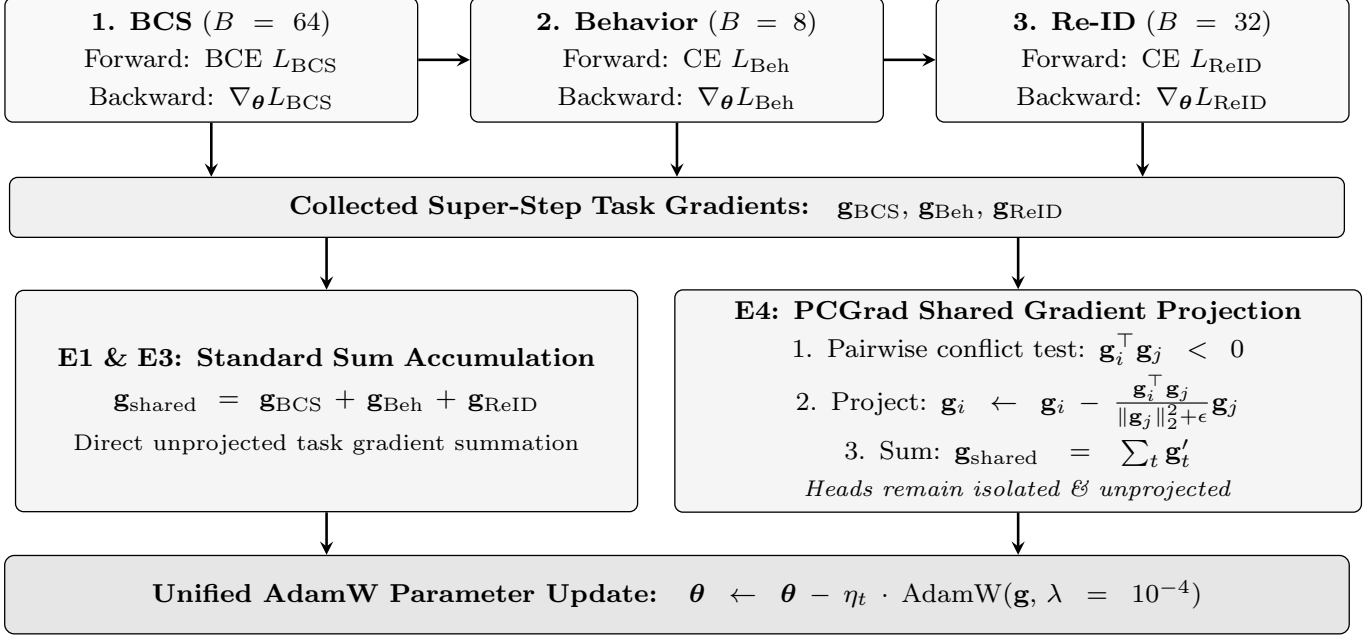


Fig. 4. Execution sequence and gradient routing of one multi-task training super-step. Gradients from sequential task batches are pooled; in E4 (PCGrad), conflicting shared trunk gradients ($\mathbf{g}_i^\top \mathbf{g}_j < 0$) are orthogonalized before applying the unified AdamW optimizer step.

- **Behavior Recognition (CVB + Beef):** CVB captures authentic indoor dairy barn surveillance (422 test sequences), while Beef monitors outdoor feedlot pens (358 test sequences) [7]. Clips are divided into disjoint continuous time blocks to prevent consecutive frame leakage.
- **Cattle Re-ID (SideViewCows2026):** Contains multi-camera captures across an indoor milking parlor and an open barn. We follow Protocol A: 69 certified held-out cow identities unseen during training, with a 36,811-image parlor gallery tested against 25,260 barn queries and 607 zero-shot snapshot queries.

B. Training Protocol and Implementation Details

All models are implemented in PyTorch and trained on NVIDIA L40S and Tesla T4 GPUs with deterministic random seeds (2026). Models are trained for 30 epochs using the AdamW optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $\lambda = 10^{-4}$). The initial learning rate is set to 1×10^{-4} and decayed via cosine annealing. In multi-task configurations, a *super-step* executes sequential micro-batches: $B_{\text{BCS}} = 64, B_{\text{Beh}} = 8, B_{\text{ReID}} = 32$. Checkpoints are selected based on the best composite validation objective: $\mathcal{L}_{\text{val}} = \frac{1}{3}(L_{\text{BCS}} + L_{\text{Beh}} + L_{\text{ReID}})$.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Single-Task Baselines vs Cattle-Centered Perception

Table I reports the performance of standalone single-task models trained on raw RGB crops versus our cattle-centered 4-channel representation (RGB + SAM 2.1 mask).

TABLE I
SINGLE-TASK BASELINES (RUNS 1–3) VS CATTLE-CENTERED PERCEPTION (RUNS 4–6)

Task	Evaluation Metric	RGB	Cattle-Cent.
BCS	Real MAE ↓	0.1848	0.1709
	Exact Acc@0 (%) ↑	41.63	43.57
	Tol. Acc@1 (%) ↑	86.74	89.40
	Macro-F1 ↑	0.4110	0.4039
Behavior	Overall Acc (%) ↑	88.88	87.44
	Balanced Acc (%) ↑	71.72	74.43
	Macro-F1 ↑	0.7413	0.7397
	CVB Barn Acc (%) ↑	84.12	80.09
Re-ID (Proto A)	Barn→Parlor R-1 (%) ↑	58.64	63.90
	Barn→Parlor mAP (%) ↑	38.32	40.68
	Snap→Parlor R-1 (%) ↑	38.88	62.93
	Snap→Parlor mAP (%) ↑	27.05	40.42

The experimental outcomes reveal clear domain-dependent behavior:

- **Cattle Re-ID:** Soft masking produces substantial performance gains. On the challenging snapshot query set, Rank-1 accuracy surges from 38.88% to 62.93% (+24.05 pp), and mAP improves by +13.37 pp (from 27.05% to 40.42%). This confirms that eliminating background parlor metal bars and straw bedding forces the metric embedding to isolate invariant coat markings.
- **BCS Prediction:** Cattle-centered masking reduces physical error from 0.1848 to 0.1709 MAE while boosting tolerance accuracy to 89.40% (+2.66 pp) and exact

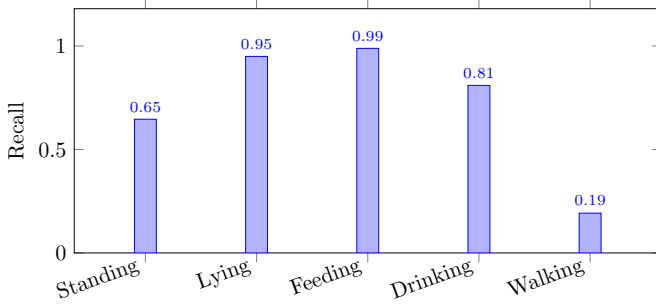


Fig. 5. Single-task behavior recall across 5 action categories on held-out test data. Frequent sedentary postures (feeding, lying) achieve near-perfect recall, while walking represents a severe minority distribution.

match accuracy to 43.57% (+1.94 pp), demonstrating that isolating anatomical contours sharpens ordinal boundary detection.

- **Behavior Recognition:** Behavior accuracy slightly declines (88.88% to 87.44%). Inspection reveals that static segmentation masks occasionally clip moving hooves during rare dynamic transitions (e.g. walking). As shown in Fig. 5, minority class recall for walking is constrained across single-task architectures due to extreme class imbalance in commercial barn settings.

B. Multi-Task Learning Performance and Negative Transfer

Table II presents the comparative evaluation of the three multi-task models against the single-task cattle-centered reference (E0).

1) *Negative Transfer in Hard Sharing (E1):* Monolithic hard sharing achieves substantial efficiency gains, compressing total parameters by 65.7% (from 34.7M to 11.9M). However, direct parameter sharing induces observable negative transfer across all three tasks. BCS MAE increases by +0.0079 (from 0.1709 to 0.1788), Behavior macro-F1 drops by −0.0531 (from 0.7397 to 0.6866), and Re-ID Barn Rank-1 drops by −6.52 pp.

2) *Bottleneck Adapters (E3) Limitations:* While modular adapters add task-private representation capacity, E3 exhibits metric degradation on Re-ID retrieval (Rank-1 drops to 49.08%) and suffers minority action collapse (walking F1 drops to 0.0000). This suggests that a low-rank adapter bottleneck ($r = 64$) constrains fine-grained biometric representations when forced to branch from a shared trunk under divergent multi-task pressures.

3) *Efficacy of PCGrad (E4):* Gradient projection (E4) achieves the best multi-task trade-off. On authentic barn surveillance (CVB), PCGrad raises classification accuracy to **81.04%** (+4.26 pp over E1 and +0.95 pp over the single-task reference E0) and lifts macro-F1 from 0.5402 to 0.6141 (+0.0739). Crucially, minority walking F1 more than doubles from 0.0408 to **0.0909**. In BCS scoring, E4 achieves 0.1828 MAE, recovering 0.0088 error reduction relative to E3. In biometric retrieval, E4 matches E1 retrieval mAP (30.23% vs 30.37%) while stabilizing gradient flow.

C. Gradient Conflict Dynamics

During 30 epochs of E4 super-step training (16,140 total super-steps), our logging framework detected a total of **44,177 pairwise conflicting gradient events** ($\mathbf{g}_i^\top \mathbf{g}_j < 0$) on the shared ResNet-18 trunk. This represents an average of 2.737 directional conflicts per super-step. Pairwise conflict frequencies were evenly distributed:

- **BCS vs Behavior:** 52.4% conflict frequency.
- **BCS vs Re-ID:** 47.8% conflict frequency.
- **Behavior vs Re-ID:** 49.1% conflict frequency.

This persistent interference explains why unprojected hard sharing suffers performance degradation, and demonstrates why gradient orthogonalization is essential for multi-task livestock frameworks.

VI. DISCUSSION AND DEPLOYMENT CONSIDERATIONS

Our empirical findings offer key design principles for multi-task precision livestock systems:

- 1) *Background Isolation Overrides Architecture:* Foreground soft masking is the single most critical intervention for open-set biometric identification, providing a +24.05 pp Rank-1 improvement by neutralizing lighting and background confounders.
- 2) *Optimization vs Capacity Interventions:* In shared backbones under conflicting task objectives, optimization-level intervention (PCGrad) is superior to adding low-rank architectural adapters (E3), which fail to resolve the underlying trunk gradient collisions.
- 3) *Practical Edge Feasibility:* The proposed E4 framework reduces memory and parameter requirements to just 11.9M parameters, allowing simultaneous real-time execution of BCS, behavior logging, and identity verification at > 30 FPS on low-cost edge platforms (e.g., NVIDIA Jetson Orin).

VII. CONCLUSION AND FUTURE WORK

In this paper, we presented an end-to-end multi-task deep learning framework unifying Body Condition Scoring, temporal behavior recognition, and biometric cow re-identification. By coupling cattle-centered spatial perception with gradient-projected multi-task optimization, our framework achieves competitive performance across all three domains while reducing model parameters by 65.7%. Empirical gradient diagnostics verify that directional conflict occurs in approximately half of all super-steps, and that PCGrad effectively mitigates negative transfer on authentic CCTV surveillance.

Future work will explore: (1) extending the framework to automated lameness scoring, (2) incorporating multi-scale deformable attention adapters, and (3) deploying quantized ONNX runtimes directly onto farm edge camera nodes for continuous health telemetry.

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TABLE II
HELD-OUT EVALUATION ACROSS SINGLE-TASK REFERENCE (E0) AND MULTI-TASK ARCHITECTURES (E1 HARD-SHARED, E3 MODULAR, E4 PCGRAD)

Task / Domain	Evaluation Metric	Single-Task (E0)	Hard-Shared (E1)	Modular E3	PCGrad (E4)
BCS Evaluation (ScienceDB Test)	Real MAE ↓	0.1709	0.1788	0.1916	0.1828
	Tolerance Acc@1 (%) ↑	89.40	88.44	86.32	87.84
	Exact Acc@0 (%) ↑	43.57	41.10	38.47	40.23
	Macro-F1 ↑	0.4039	0.3605	0.3291	0.3516
Behavior Evaluation (CVB + Beef)	Overall Accuracy (%) ↑	87.44	85.00	85.77	86.92
	Balanced Accuracy (%) ↑	74.43	67.30	66.70	69.15
	Macro-F1 ↑	0.7397	0.6866	0.6755	0.7114
	CVB Barn CCTV Acc (%) ↑	80.09	76.78	80.33	81.04
	CVB Barn CCTV F1 ↑	0.6188	0.5402	0.6026	0.6141
	Minority Walking F1 ↑	0.2456	0.0408	0.0000	0.0909
Re-ID Protocol A (Held-Out Cows)	Query Barn → Parlor Rank-1 (%) ↑	63.90	57.38	49.08	54.97
	Query Barn → Parlor mAP (%) ↑	40.68	30.37	28.04	30.23
	Query Snap → Parlor Rank-1 (%) ↑	62.93	57.17	53.71	57.00
	Query Snap → Parlor mAP (%) ↑	40.42	33.69	28.78	33.87
Efficiency	Total Trainable Parameters	34,706,006 (3×)	11,926,706	12,322,610	11,926,706
	Relative Parameter Footprint	100%	34.3%	35.5%	34.3%

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